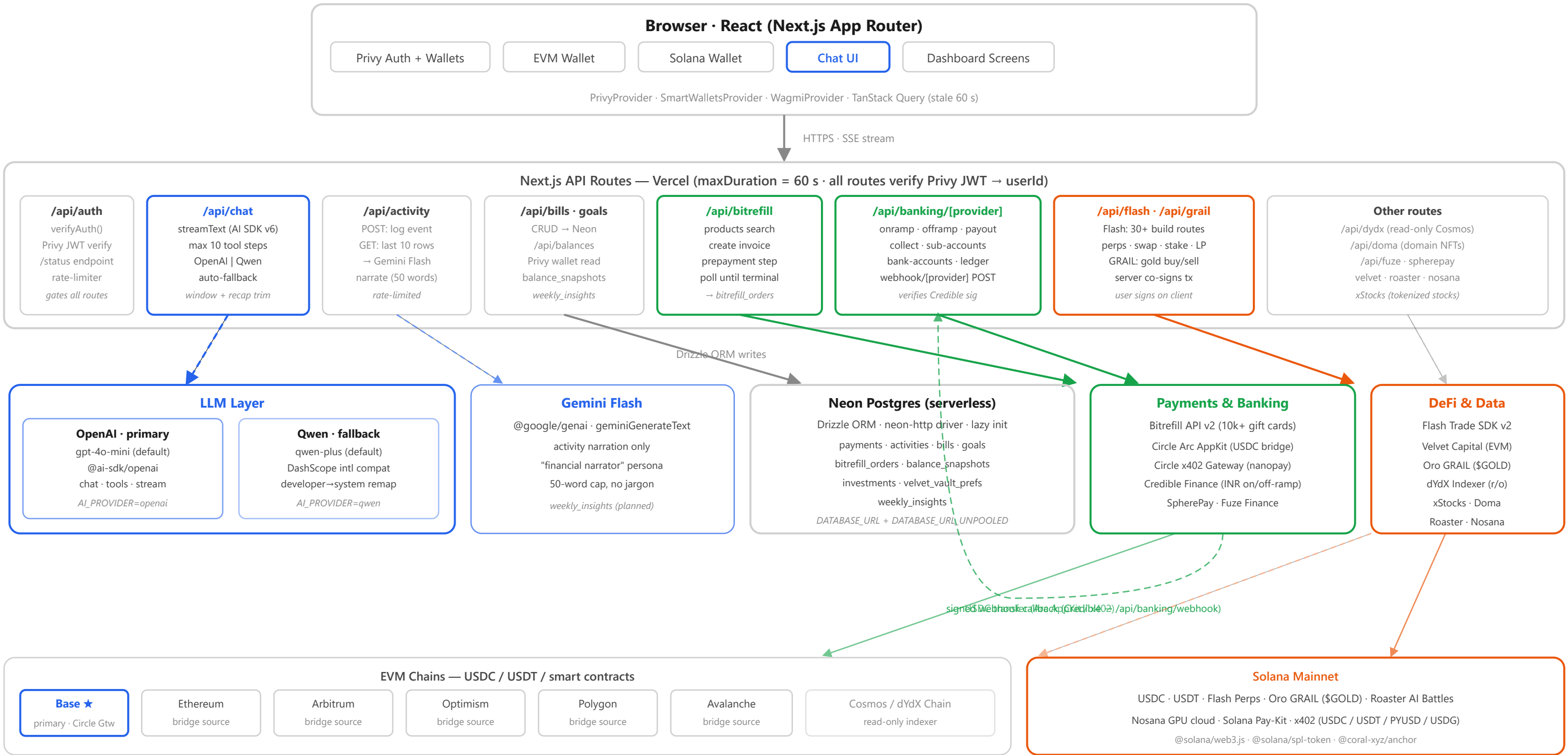


Bluvfi — System Architecture

Next.js 16 · Vercel · Neon Postgres · Privy · Circle · Bitrefill · Flash Trade · Gemini · OpenAI/Qwen



Gift card purchase — critical path (can exceed 60 s Vercel limit):

- ① User: "buy Netflix \$15" → /api/chat → OpenAI chooses buy_bitrefill_product tool call → streamText pauses, awaits tool result
- ② Tool → /api/bitrefill/invoice → Bitrefill API creates invoice → returns payment address + USDC amount → writes bitrefill_orders row (Neon, status=pending)
- ③ Arc AppKit sends USDC to address (Base) → /api/bitrefill/prepayment-step confirms send → client polls /api/bitrefill/invoice/[id] every ~5 s
- ④ Bitrefill detects payment → status → "complete" → redemption code extracted → row updated (status=complete) → LLM streams code back to user in Chat UI

INR onramp flow (async webhook):

- ① POST /api/banking/credible/onramp/order → Credible creates UPI intent → row written to payments (status=pending)
- ② User pays UPI → Credible fires signed webhook → POST /api/banking/webhook/credible → verifyWebhook() checks sig → payments row updated (status=completed, txHash)

Auth: Every route calls `verifyAuth()`, which validates the Privy JWT from cookies/headers and returns a `userId`. Rate-limiting is applied per `userId` after auth passes.

Chat: `/api/chat` uses AI SDK v6 `streamText` with up to 10 agentic tool steps. The preferred model is OpenAI (gpt-4o-mini); if `AI_PROVIDER=qwen` is set and Qwen is reachable, it runs first with automatic fallback to OpenAI. Gemini Flash is used only for the shorter activity narration endpoint.

USDC routing: Base (★) is the primary chain for payments, Circle Gateway, and Velvet vaults. Arc AppKit bridges USDC from Ethereum, Arbitrum, Optimism, Polygon, and Avalanche inward. Solana USDC runs in parallel via `@solana/pay-kit` and `x402`.

Webhooks: Credible Finance POSTs signed events to `/api/banking/webhook/credible`; the handler verifies the signature before writing to Neon — the only inbound call that bypasses Privy auth.

☐ AI / LLM ☐ Payments & Banking ☐ DeFi & Solana ☐ Infrastructure ★ *Base is the primary EVM chain* · *dashed arrows = async / webhook* · *solid arrows = direct call*